



DHSI

DIGITAL HUMANITIES SUMMER INSTITUTE

Introduction to Computational Text Analysis

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Conseil de recherches en
sciences humaines du Canada

Social Sciences and Humanities
Research Council of Canada

Canada

Introduction to Computational Text Analysis

Before you start reading

Welcome to our Introduction to Computational Text Analysis course! Below you will find some suggested readings and a list of useful links.

Soon we will send you a link to the GitHub repository for the course where we will share with you installation and homework instructions before and during the course. Our shared repository will be evolving during the course to document our activities, best reflect your interests and store any information that you will find useful.

Please make sure you have the relevant software installed on your laptop before the course starts and if you encounter any problems do write to us and we will be happy to offer guidance.

Take a look also at the website of our Computational Stylistics Group where you can find resources, method discussions and other information that may come in handy both during and after the course. While various corpora in a few languages and sizes can be downloaded from our group's website (see tab Resources there), we encourage you to bring your own corpus to the class and explore the ideas that are of most interest to you.

Looking forward to meeting you soon,

Jacek and Michał

Software Installation:

To make the most of our class time efficiently, please install the required software in advance.

If you encounter any issues during installation, feel free to contact me at: jacek.bakowski@ijp.pan.pl and I will be happy to assist you.

Below is a list of software to install. More instructions will follow soon:

- **RStudio:** <https://posit.co/download/rstudio-desktop/>
- **Python:** <https://www.python.org/downloads/>
- **Visual Code Studio:** <https://code.visualstudio.com/download>
- Here is a small collection of British Fiction novels. Please download it: https://github.com/computationalstylistics/A_Small_Collection_of_British_Fiction
- And here is a link to our **Computational Stylistics Group**: <https://computationalstylistics.github.io/>

Don't hesitate to contact me if you have some problems with your Python / Visual Code Studio / RStudio installation.

Selection of Readings:

I. Introduction to stylometry and textual analysis:

- Eder, M., Kestemont, M. and Rybicki, J. 'Stylo': a package for stylometric analyses (a manual). 2015. <https://cran.r-project.org/web/packages/stylo/stylo.pdf>

- Eder, M., Kestemont, M. and Rybicki, J. "Stylometry with R: A Package for Computational Text Analysis." The R Journal, vol. 8, no. 1, 2016, pp. 107–21. <https://doi.org/10.32614/RJ-2016-007>
- Eder, M.. "Does Size Matter? Authorship Attribution, Small Samples, Big Problem." Literary and Linguistic Computing, vol. 30, no. 2, June 2015, pp. 167–82. <https://dh2010.cch.kcl.ac.uk/academic-programme/abstracts/papers/pdf/ab-744.pdf>
- Hoover, D.L., 2013. Quantitative Analysis and Literary Studies, in: A Companion to Digital Literary Studies. John Wiley & Sons, Ltd, pp. 517–533. <https://doi.org/10.1002/9781405177504.ch28>
- Rybicki, J., Eder, M., Hoover, D. L. "Computational Stylistics and Text Analysis." in Doing Digital Humanities. Practice, Training, Research. Eds Crompton, C., Lane, R.J., Siemens, R. Oxford: Routledge, 2016, pp. 123-144.

II. Topic modelling:

- Blei David. M. „Probabilistic Topic Modelling". Communications of the ACM. April 2012, vol. 55., no. 4. <https://www.cs.columbia.edu/~blei/papers/Blei2012.pdf>
- Rhody Lisa M. „Topic Modelling and Figurative Language". Journal of Digital Humanities. Vol. 2, No. 1 Winter 2012. <https://journalofdigitalhumanities.org/2-1/topic-modeling-and-figurative-language-by-lisa-m-rhody/>
- Schöch Christof. „Topic Modelling Genre: An exploration of French Classical and Enlightenment Drama". Digital Humanities Quarterly, 11.2, 2017. <https://arxiv.org/abs/2103.13019>

III. Distributional Semantics and Cultural Studies:

- Grand, G., Blank, I. A., Pereira, F. et al. 2022. *Semantic projection recovers rich human knowledge of multiple object features from word embeddings*. Nature Human Behaviour 6, 975-987. <https://arxiv.org/pdf/1802.01241>

- Katz J., Cain Miller C., & Flynn, K. A. 2017. *The words men and women use when they write about love*. The New York Times. <https://www.nytimes.com/interactive/2017/11/07/upshot/modern-love-what-we-write-when-we-write-about-love.html>
- Kozlowski A. C., Matt Taddy, James A. Evans 2019. *The Geometry of Culture: Analyzing Meaning through Word Embeddings*. American Sociological Review, Vol. 84(5), pp. 905-949. <https://arxiv.org/abs/1803.09288>
- Miller, G. A., & Charles, W. G. 1991. Contextual correlates of semantic similarity. *Language and Cognitive Processes*, 6(1), 1–28. <https://doi.org/10.1080/01690969108406936>